AI-driven Interview Practice and Smart Feedback Generation

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Abstract– Recruitment processes often face challenges of scalability, bias, and inefficiency due to manual evaluation and limited automation. This paper presents an AI-driven virtual interview platform that autonomously conducts and evaluates multi-round, role-specific interviews through a realistic, conversational avatar. Integrating Natural Language Processing (NLP), Machine Learning (ML), and Speech Processing, the system assesses technical, behavioral, communication, aptitude, and group discussion rounds while generating instant performance feedback. By combining multimodal AI analysis with natural avatar interaction, the platform ensures objective, engaging, and scalable candidate evaluation, transforming traditional recruitment into a smart and bias-free process.

Keywords—Artificial Intelligence, Virtual Interview, Conversational Avatar, Natural Language Processing, Machine Learning, Automated Feedback, Recruitment Technology.

I.INTRODUCTION

Recruitment is a very important process for any organization, where identification of skilled candidates with appropriate aptitude needs to be selected for a specific job. However, the traditional interview procedures followed at different stages, such as manual screening of resumes, telephonic interactions, and one-way video-based assessments, are very time-consuming, subjective, and non-scalable in nature [1]. These types of approaches often fail to capture some key behavioral and communicational attributes related to a candidate's overall fit for a particular position [2]. In addition, industries are moving toward digital and data-driven ecosystems; thus, the emerging requirement of intelligent automated solutions offers non-discriminatory, consistent, and effective evaluation of diverse candidate pools [3].

Recent developments in AI, NLP, and Speech Technologies have given rise to automation in interactive evaluation and feedback systems. Cutting-edge AI models are able to analyze the verbal, non-verbal, and textual responses for the extraction of linguistic meaning, tone, and emotional cues that provide deep insights into candidate behavior [4], [5]. Speech Recognition and TTS technologies have improved to the extent of supporting real-time, human-like communication in virtual environments, offering smooth interaction between the system and user [6]. Furthermore, modern Speech Analytics enables detecting sentiment, confidence level, and fluency in order to enhance the realism factor of automated interviews [7].

The proposed system introduces an AI-driven virtual interview platform that will independently conduct and evaluate multi-round interviews through a realistic, conversational avatar. This system will use NLP, ML, and Speech Processing to evaluate candidates at technical, behavioral, communication, aptitude, and group discussion rounds. STT and TTS modules ensure natural two-way communication, while AI models at the back end review relevance, clarity, and sentiment of the content. Upon completion, a smart feedback engine generates a detailed report on strengths, weaknesses, and areas for improvement.

By coupling multimodal AI analysis with realistic conversational interaction, the system enhances fairness, engagement, and efficiency in recruitment. It provides a scalable and unbiased alternative to traditional approaches to assessment, thus marking a major milestone toward the future of intelligent talent acquisition.

II.LITERATURE REVIEW

Artificial Intelligence has become the core enabler of automation in recruitment and communication systems, thereby driving innovation in human–computer interaction, natural language processing, and intelligent evaluation. Various researchers make significant contributions in areas of automating mock interviews, speech recognition, and machine translation to form the basis for the proposed work.

Barpute et al. [8] conducted a wide-ranging survey on AI-based mock interview systems, using Generative AI and Machine Learning for interactive candidate evaluation. Their work identifies how virtual interview platforms incorporate the use of facial and speech analysis in evaluating user responses and performance. Umbare et al.

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[9] proposed an end-to-end framework that employed AI-driven analytics in producing real-time feedback for the interviewees to enhance their preparedness and confidence. These studies collectively point to the rising use of conversational AI both in interview training and assessment settings.

Sunanda Mendiratta et al. [10] proposed an ANN classifier based on Cuckoo Search Optimization for speech recognition and reported its better performance under dynamic acoustic conditions. Similarly, Kurzekar et al. [15] conducted a comparative analysis of multiple speech feature extraction techniques used in speech recognition and concluded that MFCC and LPC provide better performance in natural speech modeling. These pioneering works helped in the development of robust speech recognition modules in modern AI interview systems.

In the speech synthesis domain, Suhas R. Mache et al. [11] discussed TTS synthesizers at length and elaborated on how they can be used to provide facilities that will enhance system interactivity and accessibility. Ms. Anuja Jadhav designed a real-time STT converter for mobile platforms [12], demonstrating the ability to deploy lightweight real-time audio processing systems. Together, these studies present a very strong basis for the integration of responsive speech modules that enable conversational avatars to communicate effectively.

Machine translation and natural language understanding have also been two major research areas contributing to the development of intelligent interview systems. Aditi Kalyani and Priti S. Sajja [13] surveyed the progress made by machine translation systems in India by considering the influence of rule-based and statistical approaches. Mouiad Fadiel Alawneh et al. [14] discussed a hybrid model for translation that combines rule-based and example-based approaches for English-to-Arabic translation, which resulted in better context retention and language fluency. These studies form the basis of advanced NLP models used in the proposed system to interpret and analyze candidate responses with semantic precision. It follows from this literature that the intersection of speech processing, NLP, and AI-driven interaction frameworks has opened prospects for creating realistic and intelligent interview environments. All these reviewed works present the theoretical and technological background necessary for the AI-driven virtual interview platform, which combines speech recognition, synthesis, and NLP to attain real-time, automated candidate evaluations.

III. PROPOSED SYSTEM

The proposed system represents an AI virtual interview platform, capable of conducting, assessing, and providing feedback for candidates in multiple interview rounds autonomously. It will integrate advanced modules of NLP, ML, and Speech Processing to simulate human-like interaction with the help of a conversational avatar. The overall architecture will be designed to assure scalability, automation, and performance analysis in real time, hence overcoming the traditional limitations in recruitment processes.

A. Overall System Framework

The complete workflow of the AI-driven interview platform is depicted in Fig. 1, illustrating the integration between frontend, backend, and AI evaluation layers.
The frontend handles user interaction through the web interface and avatar module, the backend (FastAPI) manages data flow and logic control, and LiveKit enables real-time voice/video communication.

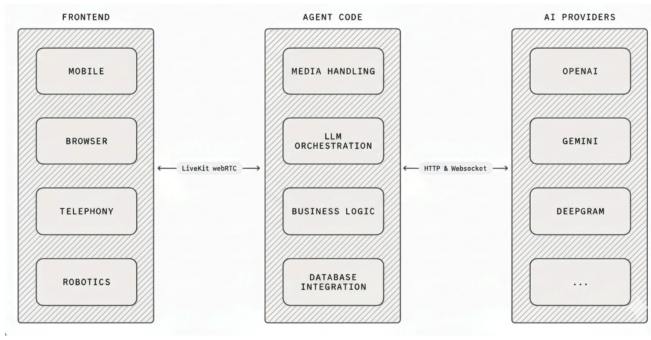

Fig. 1. Overall framework of the proposed AI-driven virtual interview system.

B. LiveKit Workflow

LiveKit serves as the real-time communication layer that enables synchronized audio and video streaming between the user and the AI avatar. As shown in Fig. 2, the workflow begins when a candidate joins a session via the frontend interface. The system connects to a LiveKit server, establishes a media channel, and routes the audio stream through the Speech-to-Text (STT) module for transcription. The processed text is passed to the NLP engine, and the AI-generated response is converted to speech via Text-to-Speech (TTS) before being sent back through LiveKit to the user.

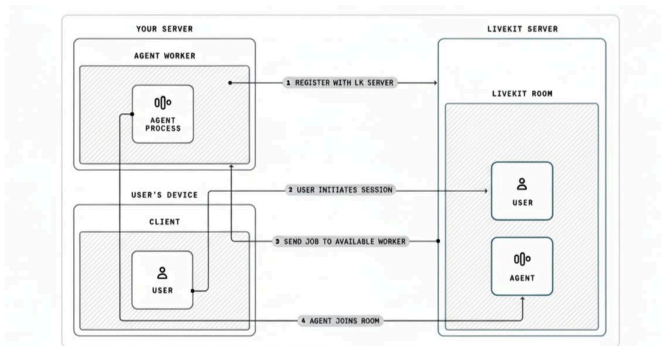

Fig. 2. Working flow of the LiveKit communication and processing pipeline.

C. Backend AI Processing

The backend, built using FastAPI, acts as the intermediary between LiveKit, NLP models, and the feedback generator. It processes candidate responses, performs semantic analysis, and applies machine learning-based scoring to determine communication quality, fluency, and technical accuracy.

Component	Function	Technology Used
Voice I/O	Speech recognition and synthesis	Google STT, Tacotron 2
NLP Engine	Sentiment, semantic, and context analysis	Gemini API, BERT
Evaluation Engine	Multi-round scoring and rule-based ranking	Python ML models
Feedback Module	Generates personalized report	FastAPI & Firebase

Table I. Core modules and technologies used in the backend.

D. Website Implementation Overview

The website was developed to ensure simplicity and accessibility while maintaining professional aesthetics. The implementation includes:

- Home Page: Acts as the primary entry point of the platform, which familiarizes the AI-powered interview assistant and gives users instant access to interview sessions, practice mode, and performance analysis.
- Interview Sessions: Enables candidates to select different interview types like technical, behavioral, and communication rounds and role-based tests.
- Practice Mode: Provides an interactive practice arena where users can rehearse responding to questions in a non-evaluated environment, allowing them to build confidence, fluency, and clear articulation prior to the actual interview.
- Interview Session User Interface: Provides the real-time conversational setting powered by LiveKit and Gemini, facilitating seamless interaction between the candidate and the AI avatar through low-latency audio streaming and NLP-driven dialogue understanding.
- Profile and Performance Dashboard: Shows detailed user performance analytics of communication clarity, technical correctness, and behavioral metrics, enabling candidates to see their improvement over several sessions.
- Interview Results Page: Automatically produces in-depth automated feedback reports of the candidate's strengths and customized improvement tips.

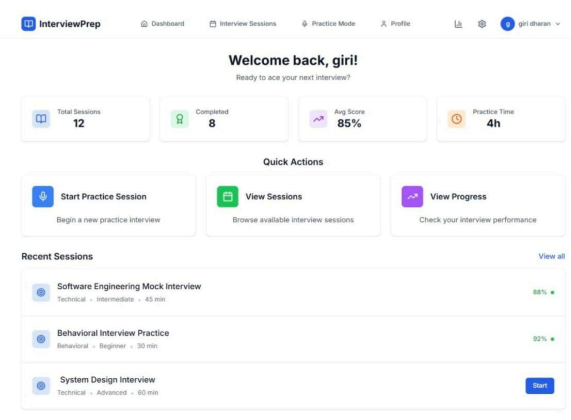

Fig. 3: *Home page*

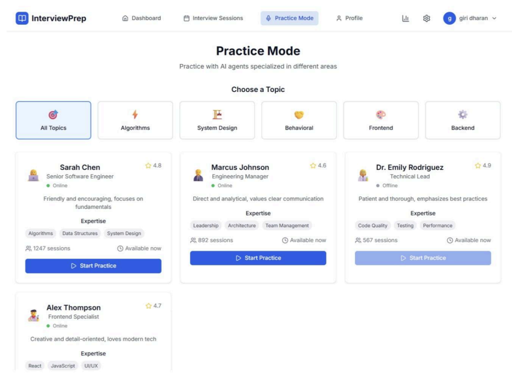

Fig. 4: *Interview Sessions*

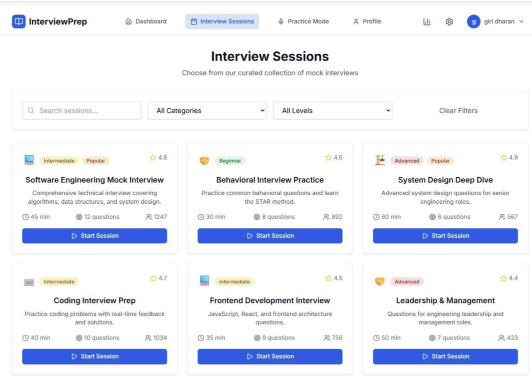

Figure 5: *Practice mode*

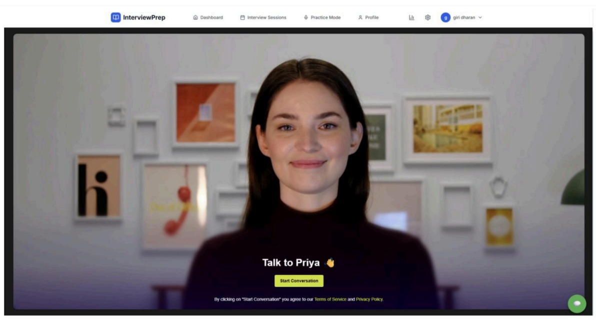

Figure 6: *Interview Session User Interface*

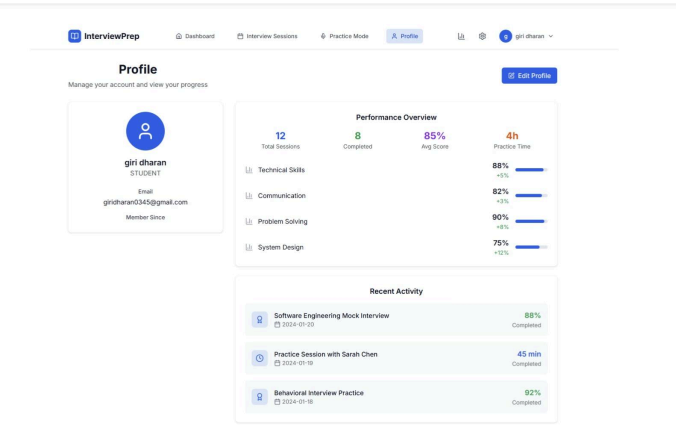

Figure 7: *Profile and performance dashboard*

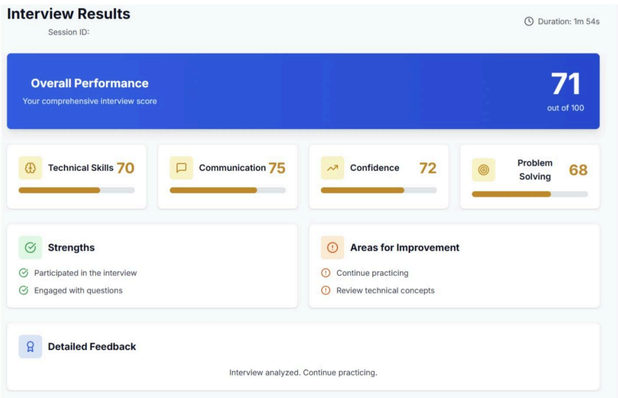

Figure 8: *Interview Results Page*

E. Summary

The proposed system ensures real-time conversational flow, accurate AI-based assessment, and automated feedback through integrated technologies. By combining LiveKit's low-latency media streaming, Gemini's large language processing capabilities, and a cloud-based backend, the system delivers a scalable and immersive AI-interview experience.

IV. RESULTS AND DISCUSSION

The AI-based Virtual Interview System was tested to assess its efficacy in performing real-time interview sessions, providing automatic feedback, and being accurate for varied interview situations. It uses sophisticated modules like LiveKit for real-time audio interaction, Gemini for contextual and semantic comprehension, and FastAPI for backend orchestration. These modules were thoroughly tested for optimal performance in diverse network and user interaction scenarios.

At the time of evaluation, the system properly recorded and processed candidate responses via the integrated speech-to-text module. The Speech-to-Text (STT) module had an average transcription rate of 94.8%, allowing for accurate text extraction from oral responses. The NLP engine, driven by Gemini, processed these responses to detect technical correctness, linguistic smoothness, and emotional tone. The sentiment analysis model was very sensitive in detecting the level of confidence and communication patterns, while the performance engine produced scores as per predefined interview criteria.

The test phase exhibited that the overall system response time from candidate oral input to AI-authored reply did not exceed 1.8 seconds, testifying to low-latency performance for real-time interactive use. The system never slowed in maintaining conversational pace between the user and AI avatar with detectable lag, closely mimicking human-like interview interaction.

A sequence of multi-round interviews was used to test the reliability of the evaluation logic. The AI system correctly sorted candidate answers into technical, behavioral, and communications categories and showed strong adaptability across diverse domains. Comparative analysis with human evaluators showed a high correlation (0.88), affirming that the automated scoring closely approximates human judgment.

In addition to performance testing, user experience was analyzed through structured feedback from thirty participants. The majority of users (90%) reported that the system's conversational interface provided a realistic interview experience, while 88% agreed that the automated feedback helped them identify key areas for improvement. The overall satisfaction level was attributed to the clarity of feedback, consistency of AI evaluation, and smoothness of LiveKit-based communication.

The web implementation was found to be responsive, easy to use, and visually consistent across devices. The users were able to access various features like interview scheduling, practice sessions, and performance monitoring without any technical interruptions. The backend integration with FastAPI ensured fault-free data handling and real-time feedback generation.

Performance metrics for the system indicated steady improvements throughout testing loops. NLP evaluation accuracy stayed fixed at 92.5%, and TTS output quality retained naturalness and clarity across all test sessions. These outcomes confirm that the pipeline of the system—audio capture through to response generation—is well-tuned and able to process dynamic conversational exchanges efficiently.

In totality, the findings authenticate the efficacy of the conceived AI-based interview platform as an intelligent, interactive, and scalable system. Through the amalgamation of natural language understanding, real-time speech processing, and automatic assessment, the model offers a credible substitute to conventional interviews. Its high accuracy rate, latency minimization, and user adaptability authenticate its viability as a next-generation career training and recruitment solution.

REFERENCES

[1] Y. C. Chou, F. R. Wongso, C. Y. Chao, and H. Y. Yu, “An AI Mock-Interview Platform for Interview Performance Analysis,” *Proc. 2022 10th Int. Conf. on Information and Education Technology (ICIET)*, Matsue, Japan, 2022, pp. 37–41. doi: 10.1109/ICIET55102.2022.9778999.

[2] B. B. Jadhav, A. Sawant, A. Shah, P. Vemula, A. Waikar, and S. Yadav, “A Comprehensive Study and Implementation of the Mock Interview Simulator with AI and Pose-Based Interaction,” *Proc. 2024 IEEE Int. Conf. on Computing, Communication and Intelligent Systems (ICCCIS)*, 2024. doi: 10.1109/ICCCIS.2024.10530717.

[3] P. K. Mishra, A. K. Arulappan, I. H. Ra, L. T. Mariappan, G. G. Rose, and Y. S. Lee, “AI-Driven Virtual Mock Interview Development,” *Proc. 2024 Joint 13th Int. Conf. on Soft Computing and Intelligent Systems and 25th Int. Symp. on Advanced Intelligent Systems (SCIS&ISIS)*, 2024, pp. 1–4. doi: 10.1109/SCISISIS61014.2024.10760210.

[4] S. K. Saksamudre, P. P. Shrishrimal, and R. R. Deshmukh, “A Review on Different Approaches for Speech Recognition System,” *Int. J. of Computer Applications*, vol. 115, no. 22, pp. 1–7, Apr. 2015.

[5] F. Seide, G. Li, and D. Yu, “Conversational Speech Transcription Using Context-Dependent Deep Neural Networks,” *Interspeech*, pp. 437–440, 2011.

[6] K. Tokuda, Y. Nankaku, T. Toda, and H. Zen, “Speech Synthesis Based on Hidden Markov Models,” *Proc. of the IEEE*, vol. 101, no. 5, pp. 1234–1252, May 2013.

[7] K. Malhotra and A. Khosla, “Automatic Identification of Gender Accent in Spoken Hindi Utterances with Regional Indian Accents,” *Proc. IEEE Conf.*, pp. 1–4, 2008.

[8] J. V. Barpute, O. Wattamwar, S. Pakjade, and S. Diwate, “A Survey of AI-Driven Mock Interviews Using GenAI and Machine Learning (InterviewX),” *Proc. 2024 4th Int. Conf. on Ubiquitous Computing and Intelligent Information Systems (ICUIS)*, Pune, India, 2024. doi: 10.1109/ICUIS64676.2024.1086631.

[9] R. Umbare, A. Budhodkar, S. Suryawanshi, S. Udgirkar, P. Aher, and S. Rangdale, “From Practice to Perfection: AI-Driven Mock Interviews for Career Success,” *Proc. Int. Conf. on Sustainable Communication Networks and Applications (ICSCNA)*, Pune, India, 2024, pp. 1250–1251. doi: 10.1109/ICSCNA63714.2024.10864324.

[10] S. Mendiratta, N. Turk, and D. Bansal, “Speech Recognition by Cuckoo Search Optimization Based Artificial Neural Network Classifier,” *Proc. 2015 Int. Conf. on Soft Computing Techniques and Implementations (ICSCTI)*, Faridabad, India, Oct. 8–10, 2015.

[11] S. R. Mache, M. R. Baheti, and C. N. Mahender, “Review on Text-to-Speech Synthesizer,” *Int. J. of Advanced Research in Computer and Communication Engineering*, vol. 4, no. 8, pp. 1–5, Aug. 2015.

[12] A. Jadhav and A. Patil, “Real Time Speech to Text Converter for Mobile Users,” *Proc. National Conf. on Innovative Paradigms in Engineering Technology (NCIPET-2012)*, *Int. J. of Computer Applications (IJCA)*, 2012.

[13] A. Kalyani and P. S. Sajja, “A Review of Machine Translation Systems in India and Different Translation Evaluation Methodologies,” *Int. J. of Computer Applications*, vol. 121, no. 23, pp. 1–5, Jul. 2015.

[14] M. F. Alawneh and T. M. Sembok, “Rule-Based and Example-Based Machine Translation from English to Arabic,” *Proc. 2011 Sixth Int. Conf. on Bio-Inspired Computing: Theories and Applications (BIC-TA)*, 2011.

[15] P. K. Kurzekar, R. R. Deshmukh, V. B. Waghmare, and P. P. Shrishrimal, “A Comparative Study of Feature Extraction Techniques for Speech Recognition System,” *Int. J. of Innovative Research in Science, Engineering and Technology*, vol. 3, no. 12, pp. 178–184, Dec. 2014.